Innovative Approaches in Child-Friendly User Interfaces: A Systematic Literature Review on Technologies, Motoric Skill and Evaluation

1st Bhanu Sri Nugraha
Dept. of Informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
bhanu@students.amikom.ac.id

2nd M. Suyanto
Dept. of Informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
yanto@amikom.ac.id

3rd Kusrini
Dept. of Informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
kusrini@amikom.ac.id

4th Ema Utami
Dept. of Informatics Doctorate
Universitas Amikom Yogyakarta
Yogyakarta, Indonesia
ema.u@amikom.ac.id

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***Abstract*—This systematic literature review examines the design and evaluation methods of child-specific user interfaces (UIs), focusing on their technological aspects, integration with motor skills, purposes of use, and evaluation methodologies. Major findings from 56 Scopus papers indicate a significant impact of UI design on children's learning effectiveness, especially through motion-based and sensory engagement. The study also highlights a novelty of innovative approaches like gamified learning and interactive storytelling, which significantly enhance user engagement and educational outcomes. Furthermore, our analysis reveals the critical role of UI in fostering social and cognitive development among children, supporting a range of learning styles and special needs. The conclusions emphasize the need for ongoing innovation in UI design tailored for children to optimize educational and developmental outcomes in digital learning environments.**

Keywords—child-computer interaction, user interface design, educational technology, motor skills development, interactive learning technologies

I. INTRODUCTION

Recent studies have explored various aspects of Child-Computer Interaction (CCI), ranging from embodied cognition to the use of digital creativity support tools [1][2]. The results of literature reviews indicate a close relationship between UI design and learning efficacy, especially in the context of movement-based learning and sensorimotor engagement [3][4]. Additionally, existing research also highlights the importance of developing interfaces that support creative expression and social engagement [5] [6].

The importance of child-friendly user interfaces in the context of environmental sustainability has also been emphasized, where well-designed technology can enhance children's awareness and participation in environmental issues [5]. Furthermore, the adoption of technology in education has expanded the scope for designing UIs that support various learning styles and educational needs of children [6]. However, despite progress in this field, much is still to be learned about how UIs can be optimized for children, including those with special needs.

The purpose of study is to conduct a novelty systematic literature review to investigate design and evaluation of UI for children. The research aims and contribution to explore research gap in various aspects of UI development and implementation specific to child users, including the technology utilized, motor skills engaged, purposes of use, and evaluation methodologies.

II. RESEARCH QUESTIONS

This study explores various aspects of UI development and implementation for children, including technology, motor skills, purposes, and evaluation methods. The research questions (RQ) are: RQ1: What technologies are used to design UIs for children? RQ2: What motor skills are involved in Children-UI? RQ3: For what purposes are Children-UI used? RQ4: What evaluation methods are used for Children-UI?

The research categorizes papers based on these questions, refined through multiple iterations, forming the data analysis foundation [7]. RQ1.1 focuses on CCI research areas, RQ1.2 on interaction stimuli, RQ1.3 on fundamental UI technology, RQ1.4 on physical materials, and RQ1.5 on digital elements. RQ2.1 addresses cognitive developmental phases, RQ2.2 identifies active body parts during UI interactions, RQ2.3 describes behaviours observed, RQ2.4 defines children's roles in research, and RQ2.5 specifies age ranges. RQ3.1 identifies skills enhanced through UI, RQ3.2 outlines study approaches. RQ4.1 explains understanding of children's states, RQ4.2 highlights interaction constraints, RQ4.3 explores abstract ideas in UIs, RQ4.4 defines cognitive processes supported, RQ4.5 specifies UI representations, and RQ4.6 discusses adult involvement in UI design.

III. BACKGROUND

Human-Computer Interaction (HCI) is an established field that examines the interactions between humans and machines, with specializations, professional associations, and significant academic presence [8].

CCI is a subfield of HCI focused on designing and implementing interactive computer systems specifically for children. CCI emphasizes creating age-appropriate, engaging, and beneficial digital experiences that children can actively participate in shaping [9][10][11]. This subfield explores the unique dynamics between children and technology, affecting areas like social skills, cooperation, and cognitive development [12].

Child motion development is essential for children's overall growth and independence. It involves the refinement of motor skills influenced by neurological and muscular development, crucial during early childhood

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[13][14][15][16][17][18]. These skills are necessary for basic movements and play a significant role in cognitive and physical growth, especially highlighted in contexts like parental interaction and formal education.

Several reviews within CCI have highlighted different aspects like the role of embodied cognition in enhancing learning through physical movement [1]. Also, the need for consistent assessment frameworks to evaluate the impact of technology on learning [19]. Otherwise, the importance of involving children in the technology design process to create inclusive and effective solutions [3]. Also calls for child-centered and ethical technology designs to ensure supportive and developmental-friendly digital experiences [4].

IV. METHODOLOGY

Review process and provide further detail on how these research questions are broken down into different categories for analysis. Table 1 show searching term on each category.

TABLE I. SEARCHING TERM ON EACH CATEGORIES

Category	Searching Term
UI Design	"User Interface Design", "UI for children", "Child-friendly UI", "Interactive Design for Kids"
Technology	"Innovative technology in UI", "Inclusive technology for UI", "Technology for child interaction", "Child-centered technology"
Motor Skill	"Motor skills in UI", "Children motor interaction", "Physical interaction in UI", "Motor development and UI"
Purpose	"Educational UI for children", "Entertainment UI for kids", "Learning interfaces", "Playful interaction design"
Evaluation Methods	"UI evaluation methods", "Children UI usability testing", "Effectiveness of children's UI", "User experience for children's UI"

To ensure the review includes relevant, high-quality literature on UI design for children, we established clear criteria using the PRISMA [20] method. We start from above keyword using Publish or Perish and using SCOPUS database only. This research doing in first quarter of 2024. The next step is explaining in some points in below.

- **Thematic Focus**: Included papers on CCI related to UI Design, Technology, Motor Skills, User Goals, and Evaluation Methods, published between 2021 and 2024. Excluded were studies outside this scope or before 2021, and non-full-length articles. This yielded 413 publications.
- **Technical & Language Aspects**: Included studies with significant technical components (over 10 citations) and in English, excluding those lacking technical focus or detailed methodology and non-English publications. This resulted in 362 publications.
- **Accessibility**: We included studies fully accessible through subscription databases, open access, or direct requests, excluding those not publicly available or requiring purchase. This resulted in 178 publications.
- **Empirical Studies**: We included studies with quantitative or qualitative data on CCI, excluding those lacking empirical data or solely theoretical. This resulted in 158 publications.
- **Finalization**: We included papers focused on HCI, excluding those purely on health, psychology, or social topics. This finalized in 56 publications.

V. RESEARCH FINDINGS

A. Technologies

To determine the percentage of subject areas in CCI research (RQ1.1), these paper cover interface design in educational games [21], child-display interaction [22], cognitive training apps for children [23], gamified learning interfaces [24], social robots for play and learning [25], safe child-robot interactions in autism interventions [26], and 3D modelling software for children [27]. Details of the scope are shown in Fig. 1.

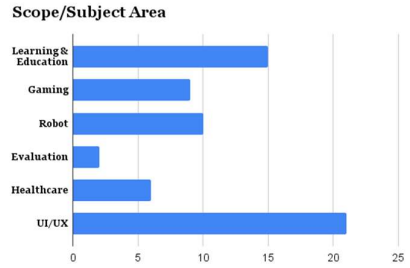


Fig. 1. Scope area of research.

The analysis shows UI Design, UI Evaluation, Education and Learning, Robotics, Health, and Games are prominent in CCI research. UI Design and Evaluation are key for educational games and applications [21][22][23][24]. Robotics research focuses on child-robot interactions, especially in education and healthcare [25][26]. Health studies examine technology in healthcare delivery and cognitive assessment for children [26][28][29]. Education research enhances learning through technology and gamification [30][31].

The range of stimuli (RQ1.2) used in interactions with children is diverse, employing various methods across different contexts, as shown in Fig. 2.

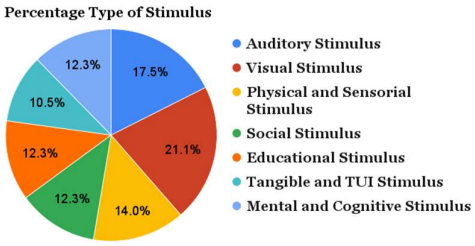


Fig. 2. Stimulus methods on research paper

References include manipulating interface design features like readiness mechanics and motion cues [21], haptic properties of robotic platforms [32], brain-computer interfaces [23], and emotion recognition in human-robot interaction [24]. Other studies cover escape room-based serious games for social interaction [33], embodied interactions with social robots [34], animistic beliefs on evaluations of therapy dogs and robots [35], upper limb assessment using natural user interfaces [36], immersive caregiver child interaction for autism [37], user experience in gamified cognitive training [38], mobile app interventions for managing anxiety [28], attention analysis in child-robot interactions [29], safe child-robot interactions in autism interventions [39], and interactive human interfaces for ERP component extraction in gifted children [40].

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The technology in UI design significantly influences children's engagement (RQ1.3). Key features include readiness mechanics, motion cues, colour cues, and rewards or penalties in educational games [21]. Wearable systems detect gait patterns [41], and Sense-based UI platforms use sensors for behavioural analysis [42]. Natural User Interface (NUI) with motion capture aids upper limb assessment in children with hemiparesis [36], and FACS improves child-robot emotional interactions [24] [43]. AI enhances UI by incorporating HCI principles, making interfaces more accessible and engaging, especially for children with disabilities [25][38][44][45].

Material choice is crucial for user engagement (RQ1.4). Tangible User Interface (TUI) with tactile materials aid visually impaired children [47]. Graphical interfaces help children understand abstract concepts [48]. Touchless gestural interfaces encourage exploration [27]. NUI and motion capture improve physical abilities assessment in children with hemiparesis [36].

Digital elements are vital in educational and healthcare applications (RQ1.5). Graphical interfaces simplify complex concepts [48]. Gamified cognitive training enhances usability for children with learning limitation [38][49]. Age-appropriate digital interfaces improve learning outcomes [45]. NUI technologies assist in rehabilitation for children with specific conditions [36].

B. *Children Motoric Skills*

The research addresses cognitive developmental phases in children's interactions with technology (RQ2.1). Studies focus on Piaget's stages: preoperational (2-7 years old) and concrete operational (7-11 years old) [27], examine cognitive load and user experience [38], highlight cognitive aspects in autism [37]. Tailored interfaces for complex subjects are emphasized [48] [47], address cognitive aspects in child-robot interactions [29], and focus on cognitive assessments for children with cerebral palsy [36].

Interactions involve various body parts (RQ2.2), including facial expressions, body movements, and sensory motor systems [25][30]. Upper limb assessments using NUIs highlight motor performance [36]. Mobile apps engage hands, fingers, and visual perception [24][33]. Voice interactions involve vocal and auditory faculties [51]. Brain Computer Interaction (BCI) engage cognitive processes [23][25].

Children's behaviours with UI (RQ2.3) show engagement, learning, and emotions. Interactions with robots boost motivation and improve behaviours [32][34]. Facial expressions in child-robot interactions affect emotional responses [24]. Voice interactions and musical games change social behaviours [51][52]. Technology improves communication in children with autism [37][53][54].

Children play key roles in research (RQ2.4), providing feedback and influencing outcomes [22][55]. Their involvement contributes to digital games and educational tools [56][57]. Involving children in design ensures technologies meet their needs, fostering engagement [58], [59]. Understanding children's evaluations of therapy dogs and robots is crucial for effective interventions [32][35].

The age range of children in this study (RQ2.5) is in Fig. 3 below.

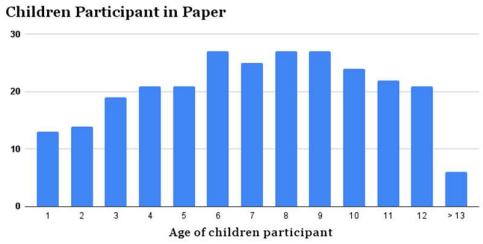


Fig. 3. Sum children involved in this study.

C. *Purposes*

Interactions with user interfaces enhance various skills in children (RQ3.1). BCI aids children with cerebral palsy in improving movement and communication [25], while interface design in educational tech promotes cognitive behaviours like stop-and-think actions [21]. Simple BCI systems enable independent movement for children with severe disabilities [23], and AR applications tailored to children's preferences enhance learning engagement [58]. TUIs help visually impaired children learn geometry through touch and hearing [47]. Child-robot interactions improve emotional connections and handwriting skills in children with dysgraphia [24][32]. Serious games foster social interaction in children with ASD [33]. Voice interactions assist caregivers in remote care management [51]. Smart mobile technology boosts creativity and social skills [56][60].

Studies used methods (RQ3.2 like eexploratory design revealed user interactions and experiences, impacting healthcare delivery [26], [53]. Usability surveys assessed user experience and cognitive load in child-focused technologies [38][61][27][31][33][48]. Observational techniques evaluated interventions like anxiety management apps and child-robot interactions [28][29]. Qualitative approaches, including in-depth interviews, provided insights into user preferences and engagement [51].

D. *Evaluation Methods*

Evaluation of user experience in gamified cognitive training for children who has learning disabilities emphasizes designing interfaces to reduce cognitive overload and enhance usability [38]. Human-robot interactions help some children with Autism Spectrum Disorder develop social skills through immersive experiences [37]. NUIs provide quantitative upper limb assessments for children with hemiparesis, aiding rehabilitation [36]. Child-cantered co-design methodology results in highly scored user experiences [63].

For data collection, evaluation methods of user experience include RedCap online surveys, descriptive analysis, co-writer setups, qualitative methods (interviews, focus groups), User-Centered Design / Human-Centered Design (UCD/HCD), and sequence diagrams in UML, cited in 7 papers. Analysis methodologies encompass thematic analysis, descriptive statistics, exploratory surveys, data mining, electroencephalogram recording, psychophysiological signal analysis, statistical analysis, and content analysis, found in 9 papers. Design methodologies include co-writer setups, UCD/HCD, sequence diagrams in UML, AR application design, anthropometric studies, the Wizard of Oz method, TUI design, and participatory design, also mentioned in 9 papers. Evaluation methods involve user experience analysis, usability testing, cognitive load measurement, observational experiments, mixed-method approaches, rapid application

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development, and usability surveys, cited in 7 papers. Additional methodologies include neuropsychological tests, daily diary studies, interactive drawing styluses, ACEs assessment, HCI features testing, wearable sensors, computer vision, METUIGA methodology, Extreme Programming, and embodied cognition theory, covered in 10 papers.

Children's UI interactions are influenced by contextual and environmental factors (RQ4.2). Voice technology enhances caregiver UI interactions [51]. UI design must account for physical factors like pressure and tilt [32][64]. Environmental constraints such as noise and distractions require adaptable interfaces [33]. Interactive homework can improve academic performance [65].

UI design should meet children's cognitive and developmental needs with easy commands and visuals [45]. Considering preferences and limitations promotes inclusivity [58]. Parental involvement (RQ4.4) is crucial for positive UI interactions [50]. Abstract concepts in UI include touchless interfaces, storytelling in education, BCIs for movement, collaborative interactions, and assistive tech for autism (RQ4.3) [27][48][23][56][62]. Externalized cognitive processes involve BCIs, interface design, assistive tech, and immersive systems for autism [23][21][62][37]. UI representations (RQ4.5) can be visualise in Fig. 4.

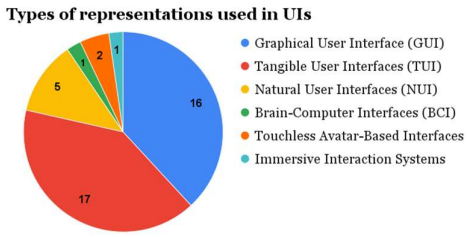


Fig. 4. Representation types used in UI.

Adults play key roles in UI design for children (RQ4.6). They must tailor UIs to children's abilities and involve them in the design process to increase engagement and effectiveness [70][71]. HCD considering children's perspectives improves UI designs and reduces risky behaviours [72].

VI. Discussion

In order to improves UI technology, some research highlights the positive impact of child-robot interaction on children's engagement, learning, and skill development, including handwriting and facial expression recognition [27] [32][24][73]. It also improves attention skills and provides assessments for cognitive impairment and hemiparesis [29] [36]. These interactions support social skills and joint attention in children with autism, similar to therapy dogs [35] [37].

Weaknesses include a lack of consideration for socioeconomic diversity [22], insufficient focus on design aspects [21], limited stakeholder perspectives [74], and inadequate discussion on long-term effects [75][76].

The impact of technology on developing and accessing fine motor skills in children, particularly those with disabilities [36], is underexplored. While some papers address rehabilitation, they lack depth on how tech interfaces target fine motor skills. Additionally, the role of interactive technology in tracking and supporting motor skills development over time is not extensively discussed.

Cultural and contextual differences in UI usability are underexplored in children's UI purpose. Ethical implications such as data privacy, security, and cognitive and emotional impacts are not extensively addressed. Literature also lacks discussion on the accessibility of UIs for children with varying abilities, emphasizing the need for inclusive designs [55][65].

In the scope of evaluating children's UI, there is insufficient discussion on the long-term effects of UI design on cognitive development [29]. Papers focus more on immediate user experience than on extended cognitive processes and knowledge retention. The significance of UI design in providing equitable access to educational resources for diverse learning needs is not fully addressed. Ethical considerations and the effectiveness of UIs in fostering collaborative learning and social interactions are also lacking.

VII. Conclusion

The crucial role of child-robot interaction in enhancing children's learning, engagement, skill development, and social interactions. Technology, especially robots, offers significant benefits for children's well-being and growth. However, these studies also highlight weaknesses, such as overlooking socioeconomic disparities, interface design issues, stakeholder perspectives, and long-term effects. Addressing these is vital for improving technology interventions for children.

The studies emphasize the importance of positive parent-child interactions, the enrichment of learning experiences through technology, and the promising outcomes of digital and immersive tools. Yet, there is a notable gap in research on using technology to enhance fine motor skills in children with impairments. Future work should focus on motion or behavior analysis, cultural considerations, and mobile technology to create inclusive and effective interventions.

Exploring motion analysis in interactive technologies to assess and improve motor skills is essential. Investigating socio-cultural influences on mobile technology design can lead to more inclusive solutions. Addressing ethical implications and accessibility in mobile interfaces will foster user-friendly and ethically sound technologies for children worldwide.

References

[1] M. Ale, M. Sturdee, and E. Rubegni, “A systematic survey on embodied cognition: 11 years of research in child–computer interaction,” International Journal of Child-Computer Interaction, vol. 33, p. 100478, Sep. 2022, doi: 10.1016/j.ijcci.2022.100478.

[2] M. H. Hagen, D. S. Cruzes, L. Jaccheri, and J. A. Fails, “Evaluating digital creativity support for children: A systematic literature review,” International Journal of Child-Computer Interaction, vol. 38, p. 100603, Dec. 2023, doi: 10.1016/j.ijcci.2023.100603.

[3] F. K. Lehnert, J. Niess, C. Lallemand, P. Markopoulos, A. Fischbach, and V. Koenig, “Child–Computer Interaction: From a systematic review towards …,” International Journal of Child-Computer Interaction, vol. 32, p. 100398, Jun. 2022, doi: 10.1016/j.ijcci.2021.100398.

[4] N. Iivari et al., “Critical agenda driving child–computer interaction research—Taking a stock of the past and envisioning the future,” International Journal of Child-Computer Interaction, vol. 32, p. 100408, Jun. 2022, doi: 10.1016/j.ijcci.2021.100408.

[5] A. Vasalou and A. Gauthier, “The role of CCI in supporting children’s engagement with environmental sustainability at a time of climate crisis,” International Journal of Child-Computer Interaction, vol. 38, p. 100605, Dec. 2023, doi: 10.1016/j.ijcci.2023.100605.

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[6] E. Eriksson, G. E. Baykal, and O. Torgersson, "The Role of Learning Theory in Child-Computer Interaction - A Semi-Systematic Literature Review," in Interaction Design and Children, Braga Portugal: ACM, Jun. 2022, pp. 50–68. doi: 10.1145/3501712.3529728.

[7] C. Robson and K. McCartan, Real world research: a resource for users of social research methods in applied settings, Fourth Edition. Chichester: Wiley, 2016.

[8] P. Markopoulos, J. Read, J. Hoÿsniemi, and S. MacFarlane, "Child computer interaction: advances in methodological research: Introduction to the special issue of cognition technology and work," Cogn Tech Work, vol. 10, no. 2, pp. 79–81, Apr. 2008, doi: 10.1007/s10111-007-0065-0.

[9] C. Sanchez, V. Costa, R. García-Carmona, E. Urendes, J. Tejedor, and R. Raya, "Evaluation of Child–Computer Interaction Using Fitts' Law: A Comparison Between a Standard Computer Mouse and a Head Mouse," Sensors, 2021, doi: 10.3390/s21113826.

[10] M. Xu, "Kid Mode on or Not? A Critical Analysis From Perspective of Child-Computer Interaction," Lecture Notes in Education Psychology and Public Media, 2023, doi: 10.54254/2753-7048/3/2022326.

[11] S. Vieira, I. Alexandre, and E. Cardoso, "The Child in Charge," 2022, doi: 10.1145/3501712.3535314.

[12] H. Chen, S. Alghowinem, S. J. Jang, C. Breazeal, and H. W. Park, "Dyadic Affect in Parent-Child Multimodal Interaction: Introducing the DAMI-P2C Dataset and its Preliminary Analysis," IEEE Transactions on Affective Computing, vol. 14, no. 4, pp. 3345–3361, Oct. 2023, doi: 10.1109/TAFFC.2022.3178689.

[13] H. Hasti, "Factors Related to the Development of Children's Motoric in Integrated Service Post Sawi Massamaturu Village, Takalar Regency," Journal of Asian Multicultural Research for Medical and Health Science Study, 2020, doi: 10.47616/jamrmhss.v1i1.19.

[14] Y. Dong, A. Aristidou, A. Shamir, M. Mahler, and E. Jain, "Adult2child: Motion Style Transfer Using CycleGANs," 2020, doi: 10.1145/3424636.3426909.

[15] K. Bigner, "Commentary on 'Effect of Active Motor Learning Interventions on Gross Motor Function and Mobility in Children Aged 2 to 6 Years With Bilateral Cerebral Palsy: A Systematic Review and Meta-Analysis,'" Pediatric Physical Therapy, 2023, doi: 10.1097/pep.0000000000001050.

[16] Y.-J. Chou, B. Y. Hu, H. Wu, C.-J. Chang, and A. Winsler, "Influence of Parental Play on Taiwanese 3-Year-Olds' Executive Function: Through the Path of Motor Skills.," Journal of Family Psychology, 2023, doi: 10.1037/fam0001006.

[17] E. C. Ningrum, N. Nurhasan, and N. Indahwati, "The Influence of the Development of Modified Teaching Materials for Kids Athletics Games on Gross Motor Learning Outcomes of Preschoolers," Competitor Jurnal Pendidikan Kepelatihan Olahraga, 2022, doi: 10.26858/cjpko.v14i2.34374.

[18] E. R. V. Dewi, H. Hibana, and M. Ali, "Loose Part: Finding Innovation in Learning Early Childhood Education," Golden Age Scientific Journal of Early Childhood Development, 2022, doi: 10.14421/jga.2022.72-01.

[19] G. E. Baykal, E. Eriksson, and O. Torgersson, "Assessment of learning in child–computer interaction research: A semi-systematic literature review," International Journal of Child-Computer Interaction, vol. 36, p. 100578, Jun. 2023, doi: 10.1016/j.ijcci.2023.100578.

[20] N. R. Haddaway, M. J. Page, C. C. Pritchard, and L. A. McGuinness, "PRISMA2020 : An R package and Shiny app for producing PRISMA 2020 ‐ compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis," Campbell Systematic Reviews, vol. 18, no. 2, p. e1230, Jun. 2022, doi: 10.1002/cl2.1230.

[21] A. Gauthier, "Manipulating Interface Design Features Affects Children's Stop-And-Think Behaviours in a Counterintuitive-Problem Game," ACM Transactions on Computer-Human Interaction, vol. 29, no. 2, 2022, doi: 10.1145/3485168.

[22] A. Schmidt, "Distance Learning, Parent–Child Interactions, and Affective Well-Being of Parents and Children During the COVID-19 Pandemic: A Daily Diary Study," Developmental Psychology, vol. 57, no. 10, pp. 1719–1734, 2021, doi: 10.1037/dev0001232.

[23] E. D. Floreani, "On the feasibility of simple brain-computer interface systems for enabling children with severe physical disabilities to explore independent movement," Frontiers in Human Neuroscience, vol. 16, no. Query date: 2024-04-08 21:35:02, 2022, doi: 10.3389/fnhum.2022.1007199.

[24] C. Filippini, "Facilitating the Child–Robot Interaction by Endowing the Robot with the Capability of Understanding the Child Engagement: The Case of Mio Amico Robot," International Journal of Social Robotics, vol. 13, no. 4, pp. 677–689, 2021, doi: 10.1007/s12369-020-00661-w.

[25] J. E. Huggins, "Challenges of brain-computer interface facilitated cognitive assessment for children with cerebral palsy," Frontiers in Human Neuroscience, vol. 16, no. Query date: 2024-04-08 21:35:02, 2022, doi: 10.3389/fnhum.2022.977042.

[26] Y. Haroun, "A human centred approach to digital technologies in health care delivery among mothers, children and adolescents," BMC Health Services Research, vol. 22, no. 1, 2022, doi: 10.1186/s12913-022-08744-2.

[27] E. Rubegni, "Child–display interaction: Lessons learned on touchless avatar-based large display interfaces," Personal and Ubiquitous Computing, vol. 26, no. 3, pp. 667–680, 2022, doi: 10.1007/s00779-020-01451-x.

[28] J. Lockwood, "Effectiveness, User Engagement and Experience, and Safety of a Mobile App (Lumi Nova) Delivering Exposure-Based Cognitive Behavioral Therapy …," JMIR Mental Health, vol. 9, no. 1, 2022, doi: 10.2196/29008.

[29] L. I. Ismail, F. A. Hanapiah, T. Belpaeme, J. Dambre, and F. wyffels, "Analysis of Attention in Child–Robot Interaction Among Children Diagnosed With Cognitive Impairment," International Journal of Social Robotics, 2020, doi: 10.1007/s12369-020-00628-x.

[30] H. Gong, "Application of Multimedia Human-Computer Interaction Technology in Preschool Children Drama Education," Advances in Multimedia, vol. 2022, no. Query date: 2024-04-08 21:17:16, 2022, doi: 10.1155/2022/6388057.

[31] B. Groba, "Stakeholder perspectives to support graphical user interface design for children with autism spectrum disorder: A qualitative study," International Journal of Environmental Research and Public Health, vol. 18, no. 9, 2021, doi: 10.3390/ijerph18094631.

[32] T. Gargot, "'It Is Not the Robot Who Learns, It Is Me.' Treating Severe Dysgraphia Using Child–Robot Interaction," Frontiers in Psychiatry, vol. 12, no. Query date: 2024-04-08 21:37:30, 2021, doi: 10.3389/fpsyt.2021.596055.

[33] G. Terlouw, "The development of an escape room–based serious game to trigger social interaction and communication between high-functioning children …" JMIR Serious Games, vol. 9, no. 1, 2021, doi: 10.2196/19765.

[34] Y. Kim, "Young children's embodied interactions with a social robot," Educational Technology Research and Development, vol. 69, no. 4, pp. 2059–2081, 2021, doi: 10.1007/s11423-021-09978-3.

[35] O. Barber, "Children's Evaluations of a Therapy Dog and Biomimetic Robot: Influences of Animistic Beliefs and Social Interaction," International Journal of Social Robotics, vol. 13, no. 6, pp. 1411–1425, 2021, doi: 10.1007/s12369-020-00722-0.

[36] C. Francisco-Martinez, "Quantitative Upper Limb Assessment With Natural User Interface in Children With Hemiparesis," IEEE Access, vol. 11, no. Query date: 2024-04-08 21:35:02, pp. 35080–35088, 2023, doi: 10.1109/ACCESS.2023.3264599.

[37] G. Nie, "An Immersive Computer-Mediated Caregiver-Child Interaction System for Young Children with Autism Spectrum Disorder," IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 29, no. Query date: 2024-04-08 21:37:30, pp. 884–893, 2021, doi: 10.1109/TNSRE.2021.3077480.

[38] A. Shaban, "Evaluation of User Experience, Cognitive Load, and Training Performance of a Gamified Cognitive Training Application for Children With Learning Disabilities," Frontiers in Computer Science, vol. 3, no. Query date: 2024-04-08 21:29:56, 2021, doi: 10.3389/fcomp.2021.617056.

[39] I. A. Katsanis, "An architecture for safe child–robot interactions in autism interventions," Robotics, vol. 10, no. 1, pp. 1–22, 2021, doi: 10.3390/robotics10010020.

[40] K. Benharrath, "Interactive Human Interface for ERP Component Extraction from Gifted Children," Intelligent Automation and Soft Computing, vol. 33, no. 2, pp. 1063–1080, 2022, doi: 10.32604/iasc.2022.023446.

[41] L. A. Boyd, "Stepping Beyond Efficacy: Understanding the User Experience of Wearables for Children with Idiopathic Toe Walking in the Natural Setting," Human Factors in Healthcare, vol. 3, no. Query date: 2024-04-08 21:29:56, 2023, doi: 10.1016/j.hfh.2023.100044.

[42] S. Lee, "Sense-based user interface platform for behavioral pattern analysis of young children," Micro and Nano Systems Letters, vol. 11, no. 1, 2023, doi: 10.1186/s40486-023-00186-7.

979-8-3315-0553-0/24/$31.00 ©2024 IEEE

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[43] N. Efthymiou, "ChildBot: Multi-robot perception and interaction with children," Robotics and Autonomous Systems, vol. 150, no. Query date: 2024-04-08 21:37:30, 2022, doi: 10.1016/j.robot.2021.103975.

[44] P. J. Cheng, "Drawing tool incorporating tangible user interface concepts enhancing children's understanding and development of color cognition," Heliyon, vol. 9, no. 6, 2023, doi: 10.1016/j.heliyon.2023.e17482.

[45] L. Giraldi, "A New Graphic User Interface Design for 3D Modeling Software for Children," Lecture Notes in Networks and Systems, vol. 240, no. Query date: 2024-04-08 12:24:41, pp. 147–154, 2021, doi: 10.1007/978-3-030-77040-2_20.

[46] J. Santos, "Home-Based Activities for Children with Speech Sound Disorders: Requirements for a Tangible User Interface for Internet of Things Artefacts," Applied Sciences (Switzerland), vol. 12, no. 18, 2022, doi: 10.3390/app12188971.

[47] R. Jafri, "Tac-Trace: A Tangible User Interface-Based Solution for Teaching Shape Concepts to Visually Impaired Children," IEEE Access, vol. 10, no. Query date: 2024-04-08 21:35:02, pp. 131153–131165, 2022, doi: 10.1109/ACCESS.2022.3228455.

[48] K. Martinez, "Graphical interface adaption for children to explain astronomy proportions and distances," Computer Science Research Notes, vol. 3201, no. 2022, pp. 119–127, 2022, doi: 10.24132/CSRN.3201.12.

[49] F. Ahmadi, "Gaming the System: A Fun Continuous Glucose Monitor Interface Improves Glycemic Outcomes for Children," Journal of Diabetes Science and Technology, no. Query date: 2024-04-08 21:35:02, 2024, doi: 10.1177/19322968231223759.

[50] M. Wahyudi, "User interface design of Be-Health application for children's learning with a gamification approach," Bali Medical Journal, vol. 11, no. 1, pp. 315–320, 2022, doi: 10.15562/bmj.v11i1.3111.

[51] E. Sezgin, "'Hey Siri, Help Me Take Care of My Child': A Feasibility Study With Caregivers of Children …," Frontiers in Public Health, vol. 10, no. Query date: 2024-04-08 21:37:30, 2022, doi: 10.3389/fpubh.2022.849322.

[52] J. A. Barnes, "Child-Robot Interaction in a Musical Dance Game: An Exploratory Comparison Study between Typically …," International Journal of Human-Computer Interaction, vol. 37, no. 3, pp. 249–266, 2021, doi: 10.1080/10447318.2020.1819667.

[53] E. Acquah-Gyan, "User experiences of a mobile phone-based health information and surveillance system (mHISS): A case of caregivers of children under-five in rural communities in Ghana," PLoS ONE, vol. 17, no. 1, 2022, doi: 10.1371/journal.pone.0261806.

[54] L. R. Ramos-Aguiar, "Teaching Emotions in Children with Autism Spectrum Disorder Through a Computer Program with Tangible Interfaces," Revista Iberoamericana de Tecnologias del Aprendizaje, vol. 16, no. 4, pp. 365–371, 2021, doi: 10.1109/RITA.2021.3125901.

[55] N. L. McElwain, "Evaluating Users' Experiences of a Child Multimodal Wearable Device: Mixed Methods Approach," JMIR Human Factors, vol. 11, no. 1, 2024, doi: 10.2196/49316.

[56] J. Sjöberg, "Collaborative interactions in problem-solving activities: School children's orientations while developing digital game designs using smart mobile technology," International Journal of Child-Computer Interaction, vol. 33, no. Query date: 2024-04-08 21:37:30, 2022, doi: 10.1016/j.ijcci.2022.100456.

[57] J. Frejd, "When Children Do Science: Collaborative Interactions in Preschoolers' Discussions About Animal Diversity," Research in Science Education, vol. 51, no. Query date: 2024-04-08 21:37:30, pp. 21–42, 2021, doi: 10.1007/s11165-019-9822-3.

[58] M. Mansor, "AR-Speaking Buddy: Designing Content and User Interface of An Augmented Reality (AR)-Based Application for .," Journal of Advanced Research in Applied Sciences and Engineering Technology, vol. 35, no. 2, pp. 57–68, 2024, doi: 10.37934/araset.35.2.5768.

[59] C. P. Wang, "Requesting Help Module Interface Design on Key Partial Video with Action and Augmented Reality for Children with Autism Spectrum Disorder," Applied Sciences (Switzerland), vol. 12, no. 17, 2022, doi: 10.3390/app12178527.

[60] A. Sultan, "Sociomaterial analysis of Azerbaijani children's smartphone use: Generational ordering …," The Emerald Handbook of Childhood and Youth in Asian Societies: Generations Between Local and Global Dynamics, no. Query date: 2024-04-08 21:17:16, pp. 213–232, 2023, doi: 10.1108/978-1-80382-283-920231014.

[61] C. Alexandrou, "User experiences of an app-based mHealth intervention (MINISTOP 2.0) integrated in …, A qualitative study," Digital Health, vol. 9, no. Query date: 2024-04-08 21:29:56, 2023, doi: 10.1177/20552076231203630.

[62] R. Cañete, "ASDesign: A User-Centered Method for the Design of Assistive Technology That Helps Children with Autism Spectrum Disorders Be More Independent in Their Daily Routines," Sustainability (Switzerland), vol. 14, no. 1, 2022, doi: 10.3390/su14010516.

[63] M. del C. del Carmen, "Reducing Dental Anxiety in Children Using a Mobile Health App: Usability and User Experience Study," JMIR Formative Research, vol. 7, no. 1, 2023, doi: 10.2196/30443.

[64] X. Wang, "Usability Design of Human-Machine Interaction Interface of Child Companion Robot in Wireless Network," Scientific Programming, vol. 2022, no. Query date: 2024-04-08 21:28:00, 2022, doi: 10.1155/2022/2840541.

[65] D. R. Skvarc, "Type of screen time and academic achievement in children from Australia and New Zealand: interactions with socioeconomic status," Journal of Children and Media, vol. 15, no. 4, pp. 509–525, 2021, doi: 10.1080/17482798.2021.1878045.

[66] J. C. Ruiz-Garcia, "ChildCI framework: Analysis of motor and cognitive development in children-computer interaction for age detection," Cognitive Systems Research, vol. 86, no. Query date: 2024-04-08 21:19:52, 2024, doi: 10.1016/j.cogsys.2024.101230.

[67] A. Q. Alban, "Heart Rate as a Predictor of Challenging Behaviours among Children with Autism from Wearable Sensors in Social Robot Interactions," Robotics, vol. 12, no. 2, 2023, doi: 10.3390/robotics12020055.

[68] S. N. Jamali, "Learning Engagement of Children with Dyslexia Through Tangible User Interface: An Experiment," International Journal of Advanced Computer Science and Applications, vol. 14, no. 11, pp. 844–854, 2023, doi: 10.14569/IJACSA.2023.0141186.

[69] B. B. Blanco, "How do preschoolers interact with peers? Characterising child and group behaviour in games with tangible interfaces in school," International Journal of Human Computer Studies, vol. 165, no. Query date: 2024-04-08 21:23:52, 2022, doi: 10.1016/j.ijhcs.2022.102849.

[70] A. C. Pires, "Learning maths with a tangible user interface: Lessons learned through participatory design with children with visual impairments and their educators," International Journal of Child-Computer Interaction, vol. 32, no. Query date: 2024-04-08 12:24:41, 2022, doi: 10.1016/j.ijcci.2021.100382.

[71] T. Nilsson, "Children's interactions with technology in teachers' self-reported activities in Sweden's preschools," International Journal of Technology and Design Education, vol. 32, no. 1, pp. 129–147, 2022, doi: 10.1007/s10798-020-09613-x.

[72] J. H. Jhone, "Is the I-PACE (Interaction of Person-Affect-Cognition-Execution) model valid in South Korea? the effects of adverse childhood experiences (ACEs) on internet gaming disorder and the mediating effect of stress on adolescents," Journal of Behavioral Addictions, vol. 10, no. 4, pp. 967–982, 2021, doi: 10.1556/2006.2021.00081.

[73] C. Provenzale, "Using Screen-Based Technologies to Assess Handwriting in Children: A Preliminary Study Choosing Human-Machine Interaction," IEEE Access, vol. 11, no. Query date: 2024-04-08 21:17:16, pp. 118865–118877, 2023, doi: 10.1109/ACCESS.2023.3326357.

[74] A. Lourenço, "Children are back to school, but is play still in lockdown? Play experiences, social interactions, and children's quality of life in primary education in the covid-19 pandemic in 2020," International Journal of Environmental Research and Public Health, vol. 18, no. 23, 2021, doi: 10.3390/ijerph182312454.

[75] J. H. Lee, "Development and Application of a Metaverse-Based Social Skills Training Program for Children With Autism Spectrum Disorder to Improve Social Interaction: Protocol for a Randomized Controlled Trial," JMIR Research Protocols, vol. 11, no. 6, 2022, doi: 10.2196/35960.

[76] N. F. Tolksdorf, "Comparing the Effects of a Different Social Partner (Social Robot vs. Human) on Children's Social Referencing in Interaction," Frontiers in Education, vol. 5, no. Query date: 2024-04-08 21:37:30, 2021, doi: 10.3389/feduc.2020.569615

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